

# FATHIMA SARA SHUMS

Junior Software Engineer — Backend & Cloud (Node.js · AWS)

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## PROFESSIONAL SUMMARY

Final-year BSc (Hons) Computer Science student (First Class Honours) with hands-on experience building backend services and cloud infrastructure on AWS. Architected and shipped AeroLink, a production-style, event-driven microservices platform (7 services) using Node.js, DynamoDB, and SNS/SQS, applying distributed-systems patterns including Saga orchestration, circuit breakers, and idempotent transaction handling. Comfortable across REST APIs, relational and NoSQL databases, Docker, and React, with additional experience in Python data pipelines from a professional data science internship. Seeking an entry-level Software Engineer role building reliable, well-tested backend systems.

## TECHNICAL SKILLS

|                            |                                                                                                                            |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Backend &amp; Cloud</b> | Node.js, Express.js, REST APIs, Microservices (Saga Pattern), AWS (DynamoDB, SNS, SQS), Docker, Docker Compose, LocalStack |
| <b>Languages</b>           | JavaScript (ES6+), Python, SQL, C#                                                                                         |
| <b>Databases</b>           | MongoDB, MySQL, PostgreSQL, AWS DynamoDB                                                                                   |
| <b>Frontend</b>            | React.js                                                                                                                   |
| <b>Testing &amp; Tools</b> | Git, Agile/Scrum, CI/CD, Unit Testing (pytest), Postman                                                                    |

## WORK EXPERIENCE

### Data Scientist Intern | Engenuity AI Feb 2025 – May 2025

- Built and maintained Python data-preprocessing and feature-engineering pipelines for client-facing ML models, working directly with real-world production datasets.
- QA-tested Power BI dashboards for accuracy and consistency ahead of stakeholder delivery, and contributed to iterative model-evaluation cycles that improved reporting reliability.
- Worked within version-controlled notebooks and reproducible experiment workflows alongside senior data scientists in a professional engineering environment.

## KEY SOFTWARE ENGINEERING PROJECTS

### AeroLink — Cloud-Native Airline Management Platform — Node.js · React · AWS DynamoDB · SNS/SQS · Docker [github.com/FathimaShums/AeroLink](https://github.com/FathimaShums/AeroLink)

- Architected a full-stack, event-driven microservices system (7 independent services) covering bookings, payments, check-in, baggage, and notifications end to end.
- Implemented a Saga Orchestrator pattern for distributed booking transactions, with compensating rollbacks on payment failure to guarantee cross-service data consistency.
- Applied production-grade reliability patterns: Opossum circuit breakers, AES-256-GCM encryption at rest for payment data, and idempotency keys to prevent duplicate charges.
- Built a live-updating React frontend with JWT-based RBAC (passenger / staff / admin), enforced on both backend middleware and frontend routing.
- Containerised all services with Docker Compose; provisioned AWS infrastructure (DynamoDB, SNS, SQS) via a LocalStack init script and a CloudFormation template for production.

### Resume Parser Pro — NLP-Powered Document Intelligence App — Python · spaCy · MongoDB Atlas · Streamlit · pytest Live: [resume-parser-ovvm.onrender.com](https://resume-parser-ovvm.onrender.com) • [github.com/FathimaShums/Resume\\_parser](https://github.com/FathimaShums/Resume_parser)

- Built a web app that parses unstructured PDF/DOCX resumes into structured data using rule-based heuristics and spaCy NLP, persisting results to MongoDB Atlas with full JSON export.
- Wrote a pytest suite covering the text-mining pipeline, field parsers, and confidence-scoring logic.

### NoSQL E-Commerce Analytics System (MERN) — MongoDB · Express.js · React.js · Node.js [github.com/FathimaShums/home-decor-ecommerce](https://github.com/FathimaShums/home-decor-ecommerce)

- Designed a full-stack analytics platform on MongoDB's document model, building 8 aggregation pipelines to power a real-time dashboard, with JWT authentication and Role-Based Access Control (Admin/Customer).

## FINAL YEAR RESEARCH PROJECT

**Adaptive Feedback Loop for Skincare Recommendation Systems** — Supervisor: Ms. Nuha Fareed, 2025–2026. Built a real-time reinforcement-signal feedback loop to improve personalisation in a recommendation engine, applying software engineering practices to a research codebase.

## EDUCATION

**BSc (Hons) Computer Science — Data Science & AI, APIIT Sri Lanka** 2023 – 2026 (Expected) • First Class Honours

